

Problem Set 14
Due: Jun. 7, 2021

1. Section 6.1 Problem 1.

Each of n packages is loaded independently onto either a red truck (with probability p) or onto a green truck (with probability $1 - p$). Let R be the total number of items selected for the red truck and let G be the total number of items selected for the green truck.

- (a). Determine the PMF, expected value, and variance of the random variable R .
- (b). Evaluate the probability that the first item to be loaded ends up being the only one on its truck.
- (c). Evaluate the probability that at least one truck ends up with a total of exactly one package.
- (d). Evaluate the expected value and the variance of the difference $R - G$.
- (e). Assume that $n \geq 2$. Given that both of the first two packages to be loaded go onto the red truck, find the conditional expectation, variance, and PMF of the random variable R .

Solution:

- (a). The random variable R is binomial with parameters p and n . Hence,

$$p_R(r) = \binom{n}{r} (1-p)^{n-r} p^r, \quad \text{for } r = 0, 1, \dots, n$$

$$\mathbf{E}[R] = np$$

$$\text{var}(R) = np(1-p)$$

- (b). Let A be the event that the first item to be loaded ends up being the only one on its truck. This event is the union of two disjoint events:

- (i) the first item is placed on the red truck and the remaining $n - 1$ are placed on the green truck,
- (ii) the first item is placed on the green truck and the remaining $n - 1$ are placed on the red truck.

Thus,

$$\mathbf{P}(A) = p(1-p)^{n-1} + (1-p)p^{n-1}$$

- (c). Let B be the event that at least one truck ends up with a total of exactly one package. The event B occurs if exactly one or both of the trucks end up with exactly 1 package, so

$$\mathbf{P}(B) = \begin{cases} 1 & \text{if } n = 1 \\ 2p(1-p) & \text{if } n = 2 \\ n(1-p)^{n-1}p + np^{n-1}(1-p) & \text{if } n = 3, 4, \dots \end{cases}$$

(d). Let $D = R - G = R - (n - R) = 2R - n$. We have

$$\mathbf{E}[D] = 2\mathbf{E}[R] - n = 2np - n$$

$$\text{var}(D) = 4\text{var}(R) = 4np(1 - p)$$

(e). Let C be the event that each of the first 2 packages is loaded onto the red truck. Given that C occurred, the random variable R becomes

$$R = 2 + X_3 + X_4 + \cdots + X_n$$

Hence,

$$\mathbf{E}[R \mid C] = \mathbf{E}[2 + X_3 + X_4 + \cdots + X_n] = 2 + (n - 2)\mathbf{E}[X_i] = 2 + (n - 2)p$$

$$\text{var}(R \mid C) = \text{var}(2 + X_3 + X_4 + \cdots + X_n) = (n - 2)\text{var}(X_i) = (n - 2)p(1 - p)$$

Finally, given that the first two packages are loaded onto the red truck, the probability that a total of r packages are loaded onto the red truck is equal to the probability that $r - 2$ of the remaining $n - 2$ packages go to the red truck:

$$p_{R|C}(r) = \binom{n-2}{r-2} (1-p)^{n-r} p^{r-2}, \quad r = 2, \dots, n.$$

2. Section 6.2 Problem 13.

Transmitters A and B independently send messages to a single receiver in a Poisson manner, with rates of λ_A and λ_B , respectively. All messages are so brief that we may assume that they occupy single points in time. The number of words in a message, regardless of the source that is transmitting it, is a random variable with PMF

$$p_W(w) = \begin{cases} 2/6 & \text{if } w = 1 \\ 3/6 & \text{if } w = 2 \\ 1/6 & \text{if } w = 3 \\ 0 & \text{otherwise} \end{cases}$$

and is independent of everything else.

- (a). What is the probability that during an interval of duration t , a total of exactly nine messages will be received?
- (b). Let N be the total number of words received during an interval of duration t . Determine the expected value of N .
- (c). Determine the PDF of the time from $t = 0$ until the receiver has received exactly eight three-word messages from transmitter A.

- (d). What is the probability that exactly eight of the next twelve messages received will be from transmitter A?

Solution:

- (a). Let R be the total number of messages received during an interval of duration t . Note that R is a Poisson random variable with arrival rate $\lambda_A + \lambda_B$. Therefore, the probability that exactly nine messages are received is

$$\mathbf{P}(R = 9) = \frac{((\lambda_A + \lambda_B)t)^9 e^{-(\lambda_A + \lambda_B)t}}{9!}$$

- (b). Let R be defined as in part (a), and let W_i be the number of words in the i th message. Then,

$$N = W_1 + W_2 + \cdots + W_R$$

which is a sum of a random number of random variables. Thus,

$$\begin{aligned}\mathbf{E}[N] &= \mathbf{E}[W]\mathbf{E}[R] \\ &= \left(1 \cdot \frac{2}{6} + 2 \cdot \frac{3}{6} + 3 \cdot \frac{1}{6}\right) \cdot ((\lambda_A + \lambda_B)t) \\ &= \frac{11}{6}(\lambda_A + \lambda_B)t\end{aligned}$$

- (c). Three-word messages arrive from transmitter A in a Poisson manner, with average rate $\lambda_A p_W(3) = \lambda_A/6$. Therefore, the random variable of interest stands for the 8th arrival time of the Poisson process, whose distribution is Erlang of order 8. Its PDF is given by

$$f_X(x) = \frac{(\lambda_A/6)^8 x^7 e^{-\lambda_A x/6}}{7!}$$

- (d). Every message originates from either transmitter A or B, and can be viewed as an independent Bernoulli trial. Each message has probability $\lambda_A/(\lambda_A + \lambda_B)$ of originating from transmitter A (view this as a “success”). Thus, the number of messages from transmitter A (out of the next twelve) is a binomial random variable, and the desired probability is equal to

$$\binom{12}{8} \left(\frac{\lambda_A}{\lambda_A + \lambda_B}\right)^8 \left(\frac{\lambda_B}{\lambda_A + \lambda_B}\right)^4$$

3. Suppose a bank has 2 tellers. When you enter the bank, you find that both tellers are busy serving other customers, and there are no other customers in queue. The service times of the two tellers are independent distributed exponential random variables, with parameters λ_1 and λ_2 respectively. What is the probability that you will be the last to leave?

Solution: We can split the whole problem into two stages.

At the first stage, you wait until one of the two customers leaves. Denote that the customer

served at the first teller A and the other B. The probability that A leaves first is $\lambda_1/(\lambda_1 + \lambda_2)$. It is because that we can view the process as a merge of two Poisson processes with parameters λ_1 and λ_2 respectively. Therefore, the probability that A leaves first is just the probability that the success of the merging process belongs to the first one, i.e., $\lambda_1/(\lambda_1 + \lambda_2)$.

At the second stage, you and the other customer will be served. Suppose that A leaves first, then you are served at the first teller. Due to the memoryless property of exponential RVs, the probability that B leaves earlier than you is $\lambda_2/(\lambda_1 + \lambda_2)$.

Therefore,

$$\begin{aligned}\mathbf{P}(\text{you leave last}) &= \mathbf{P}(\text{A leaves first})\mathbf{P}(\text{you leave last} \mid \text{A leaves first}) \\ &\quad + \mathbf{P}(\text{B leaves first})\mathbf{P}(\text{you leave last} \mid \text{B leaves first}) \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2} \cdot \frac{\lambda_2}{\lambda_1 + \lambda_2} + \frac{\lambda_2}{\lambda_1 + \lambda_2} \cdot \frac{\lambda_1}{\lambda_1 + \lambda_2} \\ &= \frac{2\lambda_1\lambda_2}{(\lambda_1 + \lambda_2)^2}\end{aligned}$$

4. ***Bonus.** Customers arrive at a certain retail establishment according to a Poisson process with rate λ per hour. Suppose that two customers arrive during the first hour. Find the probability that both arrived in the first 20 minutes.

Solution: Let $N(t)$ denote the number of customers arrived in the first t hours. We have $\mathbf{P}(N(t) = k) = e^{-\lambda t}(\lambda t)^k/k!$.

$$\begin{aligned}\mathbf{P}(N(1/3) = 2 \mid N(1) = 2) &= \frac{\mathbf{P}(N(1/3) = 2, N(1) = 2)}{\mathbf{P}(N(1) = 2)} \\ &= \frac{\mathbf{P}(N(1/3) = 2, N(1) - N(1/3) = 0)}{\mathbf{P}(N(1) = 2)} \\ &= \frac{\mathbf{P}(N(1/3) = 2) \cdot \mathbf{P}(N(1) - N(1/3) = 0)}{\mathbf{P}(N(1) = 2)} \quad (\text{independence}) \\ &= \frac{\mathbf{P}(N(1/3) = 2) \cdot \mathbf{P}(N(2/3) = 0)}{\mathbf{P}(N(1) = 2)} \quad (\text{time homogeneity}) \\ &= \frac{\frac{e^{-\lambda/3}(\lambda/3)^2}{2!} \cdot e^{-2\lambda/3}}{\frac{e^{-\lambda}\lambda^2}{2!}} \\ &= \frac{1}{9}\end{aligned}$$